

Good afternoon, so time for the last module in the course, so I change the background again, some might recognize this one, we'll see.

This is the last module we have, module seven on quantum mechanics, it's an introduction to quantum mechanics, so I'm not, there's whole courses on quantum mechanics which I'm not gonna cram into one and a half weeks, just a flavor of what it entails, but still we can do some nice things with that already.

Let's first have a look at the recap question of a previous time, so I displayed this already, you may have given it a thought, so please take a seat guys, so let's see if you can make something of this.

This was an exam question, multiple choice question, I think two years back or in the recent I've got, what you don't have now, probably, but what you would have at the final test is the formula sheet, so that contains this formula, which isn't much of a help if you don't know how to apply it, but may just give you the edge to solve this, so let's vote who thinks answer A.

Answer B comes larger, answer C comes smaller, answer D, there's no longer critical.

Answer B is correct, so what happens?

So there's a lot of information in the question, so I'm testing your reading skills again, so we had quartz glass with this dispersion curve, we're going to change that to fluoride, which is this one, and we were at what is it 500 nanometers, and we're going to increase that to 650 to here, so we go from this refractive index to this refractive index, right, that means that n is going down, so and if you think about formula that you have for the critical angle, so n_B , this is one error because critical angle only happens if you go from a medium with a higher refractive index to a medium of lower refractive index, and then n_A is decreasing because it was previously, well let's say one point, let's call it 1.55, and then now it's going to, well less, well around 1.43, so n_A is decreasing, so the fraction becomes larger, so the critical angle has to increase, right, is that clear? Okay, good, so what we looked at last Monday was dispersion, polarization, bruiser angle, also scattering and interference in the fraction, and especially the latter two I'll get back to today, because this is also relevant for quantum mechanics, so today I'm mainly covering chapter 38, and there might be a few changes here, I might include the harmonic oscillator as well, but I'll update you next week on that, but today is mainly chapter 38, and there's a bit of history involved, this is just for reference, you don't have to recall this, so at the end of the 19th century they got pretty much figured out all of physics up to then, so they had Maxwell's law, this was verified experimentally by Hertz composed, and physics was capable of explaining most of this, just a few minor questions, they thought just as with light, just as with water waves, that these waves had needed a medium to propagate through, so water waves propagate through water, they thought that light waves also, or light itself, had to propagate through some kind of medium called ether, and there were experiments that tried to measure that, and then these experiments proved to be inconclusive, they could not measure that there was ether around, and that led to the general theory of relativity, and also to the fact that light could propagate also through vacuum, which was weird, especially at that time, and also the atom was not completely understood, they could not explain the photoelectric effect, which I'll come back to in a second, or in a few minutes actually, and they had some basic idea of how atoms were built, but not entirely, there was an experiment by Werther Ford, who showed that atoms had a nucleus, but around that, yeah, electrons should be going round, but if they were going round, why did they not fall back to the core, because the nucleus positive, electrons negative, they should attract, so they could not explain how that was happening, and it took from then on around 30 years to build quantum mechanics, initiated by

Einstein, not a fan himself, but he still contributed quite a bit to the photoelectric effect for instance, and this then developed into quantum electrodynamics, and then quantum chromodynamics and the standard model, as we you've probably seen it taught in secondary school, there's also some efforts to unify the forces, I will not go in detail here, and the standard model is also still under development, and this is really high energy physics that contributes to that, mass of neutrinos, the Higgs boson discovered in 2012, all of that is just a history that has led to where we are now, well general relativity was also verified, this led to a Nobel prize, the big bang theory was understood, but there are still problems with what happens in the universe, a lot of things, a lot of mass in the universe is not accounted for, we don't know where it comes from, how it can be explained, if you look at how the universe is expanding, how it is twisting and turning, we need more mass to explain that, and that results to theories on dark matter, we can't measure it, it's invisible dark matter, and also dark energy is coupled to that, so a picture of NASA shows that just a small part of the universe as we know it now is explained, this dark energy dark matter which composes most of the universe, I will not go in depth in this, just to zoom out again, how is this and why is this relevant for you as an electrical engineering student or a master AI and ES student, psychology and technology also right, I'm not sure, I think we had a few of those as well, we need quantum mechanics to understand charge transport, this will link ultimately back to module one, although I will not go in depth in how energy levels and band gaps are formed in atoms, it's just that the atomic structure is the base for what happens with the band gaps in semiconductor physics, and some founding principles are useful and that's why they are included in this module, and of course for the field of quantum computing this is a key element, quantum entanglement is really something, I'll conclude with that next Thursday, that's really something that is weird to understand but that's really an effect that takes place, so let's see why just what we had up to then was not sufficient to explain things that we measured, so we know that light and that's what I call for the module six can experience wave-like behavior interference and diffraction, those are really things that can be explained with the wave nature of light, but there's also a particle behavior of light, photoelectric effects and photons are then used which is the quantum of energy and light, those are used to describe the quantum nature of light, and the rest of this lecture will go about, well I will talk about the quantum nature of light and how to explain that and how to well some things cannot be explained, I just have to tell you how they are and there's no why they are like that, but the term quantum is a discrete quantity of energy, the smallest amount of energy that you can subdivide anything into, and as we'll see this relates to the frequency of light, proportional to the frequency of light and the proportionality constant is a very very small number called Planck's constant, so but before we get there let's first look at the photoelectric effect, so what happens if light shines onto an object, certain materials can emit electrons if there's sufficient light shining on them, so this material has atoms and there's electrons in the material and such an electron can break free from the material if it gets enough energy supplied by the light, so that's the photoelectric effect, so you provide energy to the material and then electrons can escape from the material and they do this with a certain kinetic energy, they can have a maximum velocity and then the maximum kinetic energy they have is half times the mass of the electron times the kinetic energy squared, sorry the velocity squared that builds kinetic energy, but we're talking about really really small numbers, so the electron has to absorb this energy to overcome this attractive force that is inherent to the material and that's called the work function ϕ , so there's an intrinsic energy property of the material that we have to overcome to be able to have electrons escape and this light supplies a kick to kick these electrons free, right, so how does that look like if you do an experiment, this is done with a vacuum tube, so there's vacuum in the tube and there's a large electrode and a wire electrode and you shine light on this electrode and depending on the polarity one of them is a cathode, one is the anode, in this case you put the larger one, the larger electrode as the cathode, the negative one and this is

positive and then you make an electric field from the anode to the cathode, if you then shine light on this cathode and the light has enough energy you can break free these electrons and they will accelerate according to the electric field, so this is understandable, so that's what the electric field does and this force is then, well the charge times the electric field and the electric field depends on the voltage you apply across the two electrodes and this then gives an acceleration to the electron of mass m_e , so the electron will accelerate so the kinetic energy will increase and over time it will go to a higher velocity, so it will speed up as soon as it's released from the cathode, so this is how you operate it in forward motion and forward bias, if you would reverse the electric field so if you would reverse the anode and cathode so this becomes the anode that the cathode then you will reverse the electric field, so any electron that becomes free, it gets freed from the surface here will be attracted again by the anode, electrons go to the positive pole to the anode, that might be a typo then sorry this is the cathode, this is the anode, yeah so the electron will go back to the to the anode, I'll fix that, after a certain potential if you apply a certain potential to this there are no electrons that reach here anymore, so all of them who get freed by the light shining on it will go back to the same electrode and this is called the stopping potential V_0 , this happens at a certain voltage, and if you look at how much work has been done there then the work done by the other forces in this case the electric field, work is done it's calculated by the charge times the voltage that the charge undergoes, so that's e the electron charge times the voltage then relates to the change in kinetic energy, so if it comes to a full stop at this electrode then all of this kinetic energy has been converted into work and at that voltage no electrons reach this point anymore, if you observe this experimentally then the current you get depends on the frequency, it depends on the light frequency, not on the amount of electrons well sorry, the intensity does matter but it happens at a

certain frequency that this effect starts to take place, there is no real time delay, nothing measurable between when the light hits the surface and when the electrons are released, so this happens instantaneous and the stopping potential does not depend on the intensity of the light, only depends on the color of the light, the frequency of the light, so these are effects that cannot be explained with wave behavior of light, so we need something else to describe this, well to summarize in a different way if you have higher intensity of light you do get more photons, more quanta of light hitting the material and that would lead in a higher intensity for a positive voltage but for a negative voltage the stopping potential does not depend on the intensity of the light, just on the color of the light, so the wavelength of the light or the frequency of the light, so if you give more intensity you get more photons, sorry more electrons release so you get a larger current, so if you double the intensity you get twice the current but still the stopping potential does not change, this depends really on the frequency of the light and that is weird behavior, it should, they should both change if you think of a wave description of light but that's not the case, the only way to explain this and that's what Einstein did is to link this to a quantum description of electrons and photons, so he had to introduce a quantum of energy that was needed to knock these electrons free from the material and that's his explanation of the photoelectric effect that led to quantum physics, so he postulated in 1905 and it was just at the start of the discovery of quantum physics that the energy of light should be quantized in, should be localized in small bundles, the smallest amount is a photon, so a quantum of light and the energy of such a photon and this is an important formula that's also in a formula sheet, the energy is proportional to the frequency of the light times the proportionality constant, Planck's constant, that will appear in a second here, we also know that frequency is c over λ , so there's many different ways of writing this, hf or hc over λ is the two most common ways of seeing this formula, so if you incorporate this in what we had earlier for the photoelectric effect then this energy was supplied to the cathode or to the to the electrode and then it would knock free electrons, so it had to overcome the work function that we had and would give the electrons a certain

kinetic energy, the maximum kinetic energy then related to well the charge times the voltage that the electron underwent, so phrased in a different way, if we put k_{max} as the work done on the electron we get $e \times v_{\text{naught}} = hf - \phi$ and that is a curve that scales linearly with frequency, so for a higher frequency we get a higher kinetic energy, so a higher work done on the electron and this proportionality constant, the slope here, that's Planck's constant and that's a very very small number, so we're talking about light and photons, so the smallest amount of energy contained relates to a really really small number times the frequency, so even if you think of terahertz which is 10^{12} in frequency then you still have a very small amount of energy, $h \times f$ is still very small, there's a question here, oh it should be $e \times v$ actually, here on the vertical axis I may have copied the wrong figure, really the slope is typically depicted as Planck's constant, I may have put the wrong label, I'll check that, it should be $e \times v$ probably because the slope is typically really denoted as Planck's constant, yeah thanks for remarking that, I'll check that, often you see it more stylized, you have the work function divided by e indeed, so this is, yes it is a scaled Planck's constant, actually the answer is here, so the slope relates is proportional to Planck's constant, I should have mentioned that, it's h/e , so it's correct, yeah okay, solved, if you have a different material, such a material has a different work function, so certain materials have a work function which is expressed in electron volts, which is a useful unit because that's the same as the left hand side, depending on the material you end up at a different intercept on the y axis and that then relates to the stopping potential scaled with the electron charge, so this is the curve depicted, so it's indeed h/e , that determines the slope, but since e is a constant, well it's a way to calculate h basically, are the questions so far, yeah sorry, ϕ is the work function, so let me get back to here, I don't mind saying that again, yes, so to kick free an electron from a certain material, you need to overcome certain energy that the material has intrinsically, so I need to kick these electrons too, so they can escape and that relates to the type of material that you have, if you have larger atoms then you need less energy for instance to kick this free, if you have smaller atoms then you need more energy and this relates to, ϕ is an energy that you need to kick free electrons, all right, yeah then no not completely, because the band gap is just to have electrons go from bound state to freely roaming state within the material, this is really to kick them free from the material, so they escape the material, so it's a similar type of measure metric, but while you free the electrons completely instead of just having them roam rounds in the material, it's a combination of both probably, just to get it across the band gap and then kick it free from the material, I think the band gap energy is much less than typically what you need as work function to get the electron really out of the material, so does that explain the work, okay good, so where were we here, so the interpretation of the photoelectric effect then is that if the energy of the light is not enough, so less than the work function, less than ϕ , photons cannot release electrons from the material, so don't have enough energy to kick them free, and this means that there is a certain minimum frequency, since h is a constant there is a certain f_{critical} , a minimum frequency that you need the light to have to kick free the electrons, and this is really that there is such a minimum bound that's really an effect that relates to quantum mechanics, that's really a quantized form of energy that you need to accomplish anything within the material, and this process is instantaneous, so it's a particle collision between the photon and an electron or the material that has the electron and that gets kicked free, and if you fix the frequency there is a certain stopping voltage at which this effect does not happen anymore, and that's something you can measure for any material, that's a way to determine this work function, and then you measure curves like these for various frequencies of light, you measure points hopefully on a line, and then you look at the intercept to find the work function, that's how these work functions have been determined, question?

Yeah it's more of an interaction indeed, yeah I'll get back to the content effect,

content scattering that kind of explains this as well, there is conservation of momentum, there is conservation of energy, but it's done in a weird way, I'll get back to that later, I cannot answer that now because it disturbs the flow in the slide, I'll get back to that. So the bottom line of this, of the photoelectric effects from the experiments, is that we now know that light behaves kind of like a stream of particles, and these particles are called photons which have quantized energy, so they have discrete levels of energy that we can let material interact with, so please try to summarize this for yourselves in two minutes, and then we'll continue with another effect that happens here, it is weird to wrap your head around, I have to admit, please try to write it down, so let's see what you can make of a quick quiz question, let's see if you understand the concepts, so this is really, if you think about this as a collision, then we're talking an absorption, which you could consider a completely inelastic collision, and then something happens, a follow-up, so an electron is absorbed, and then it's, sorry, a photon is absorbed, and then an electron is released, but let's see if you can, if you understand the concept, so two blanks, four options, and one option that says none of your more, let's vote, who thinks answer A, answer B, answer C, answer D, and none of the above, answer B is indeed correct, the color of the light determines the energy, but if you have more of the same light, you get more electrons emitted per second, but the maximum kinetic energy does not change, so that relates to the frequency of light that you shine on the electrode, so two other effects that could not be explained with just a particle behavior of light, so yeah, with a wave behavior of light, but need a particle behavior, are x-ray production and content scattering, so to go through those, so we had the photoelectric effect that showed photon absorption, we can also do it the other way around, we can shoot electrons at an anode, and then light can be released, so we need for this a heated coil that can emit electrons, and then if there's a potential difference between this part and that part, the electrons will accelerate from the cathode to the anode, so they will accelerate against the electric field, that's how they are attracted, they will hit the anode, and then they can emit, due to this collision, light can get emitted, so you are bombarding a surface with fast-moving electrons, and then you generate radiation, and this was done in 1895 by Röntgen, who discovered this, and what happens if you shoot electrons at this, and then there's a whole spectrum of light that can get emitted, depending on how the electron gets absorbed, can be absorbed all at once, or it can be absorbed in stages if it has multiple collisions, and then you get a spectrum of radiation that comes out, but I'll show what that looks like in the next slide, so we need a high voltage to do this, kilovolts, and the anode slows down the electrons quite abruptly, but it can happen all at once, but also over a small time instant, and in German this is called Bremsstrahlung, breaking radiation, Röntgen was a German guy, and these deceleration of the electrons, so they break down, and any accelerating or decelerating electron will generate electromagnetic radiation, electromagnetic waves, and this comes out of the x-ray.

This is typically at a very very small wavelength, much smaller than visible light, so ranging from nanometers to picometers, 10 to the ninth to 10 to the 12 minus 12, sorry 10 to the minus ninth to 10 to the minus 12, and this gives a continuous spectrum of EM radiation, however not completely continuous, because the wave model would predict that you get a complete continuous spectrum, but if you do measurements on this, then there is a certain minimum wavelength that you get for a certain voltage that you apply to the between the cathode and anode, so the wave model would predict something continuous over all wavelengths, but measurements show that there is some clear lower threshold for wavelength, the spectrum spectra that are measured experimentally have a minimum wavelength or a maximum frequency, those are interchangeable, depending on the voltage that you apply, so we get a shorter minimum wavelength if you apply a higher voltage. AC does not stand for the alternating current here, but just the voltage between anode and cathode, so this is a DC voltage, and the wave model cannot predict this, a photo model can, and the reason is as follows, this electron of charge minus e undergoes this potential

change, so there's work done on the electron that has a value of E times this potential, and that is then converted into kinetic energy, if this is, sorry, this kinetic energy relates from the voltage, and if this kinetic energy is used to bring this electron to a full stop, all of this energy is converted into a photo, and that has a certain maximum frequency, so this due to this being a finite number, this being something you prescribe, there's a finite value for k that you achieve, so there's a finite value for f_{max} that you will get, and this gives, well, the work done on the electron being h times f_{max} or hc over λ_{min} , so there's a minimum wavelength or a maximum frequency that you achieve while doing this, and that relates to, well, this minimum on these experimental curves, these are idealized, of course, would be more noise if you do measurements, but there's a certain minimum wavelength that you can get from this, and that only works if you think about photons, if you think of this as photons, there is a quantized amount of energy that you can deliver to a photon, and that relates to a minimum wavelength, so this is used in x-ray imaging, and how does that work?

Well, these electrons of atoms, it also works the other way around, electrons of atoms can absorb x-rays, so then you have to process the other way around, and if you have atoms or materials that have a lot of electrons, then it's more easy for them to absorb x-ray, and that's why different types of materials, bones, they have a phosphorus, and they have calcium in them with 15 or 20 electrons, so it's easy for them to absorb x-ray, and that's why they look white in an x-ray picture, and soft tissue is mainly carbon and hydrogen and oxygen, which has less electrons, so it's much more difficult for those atoms to absorb x-ray, and that's why they give a different contrast in the picture.

Is that clear? Okay, last, okay, sorry, that's here, so there's different contrasts between the different types of tissues because of the different atoms involved.

Now we get to momentum and collisions, so we look at absorption, we looked at emission, if we look at scattering, that's a third category we can consider for photons, and if it's really scattering, that means that you absorb it and re-radiate it.

If that would happen to a photon, incident on an electron, the electron would absorb the photon and re-emit the photon, that would have the same frequency and wavelength, because that's the same amount of energy. However, if you do measurements on this, then there is scattering at a certain angle, and the wavelength is different, the wavelength is lower, so we cannot explain this with absorption and re-radiation, we need something else to explain this, and also the electron will get the momentum in a certain direction. So that's a last category of effects that we consider that cannot be described with a wave-like model. So first, oh sorry, did I say this already?

Yeah, so the wave model would do this, this is not what happens, the scattered light has a lower frequency, so a longer wavelength.

This means that if there's collisions involved, then photons should have momentum, and that relates to your question. If you want to describe this as collisions of particles, or let's call them particles, and we know that we have Einstein's quantized energy consideration, $E = hf$, then we can look at, we can define if photons have a momentum.

So also from relativity, $E = mc^2$, that's the formula you probably know, this holds for a particle that is at rest, particle with mass m , that is at rest.

A photon does not have mass, so this does not hold for a photon.

In general, how you approach this is that if the particle is not at rest, but

moving at a certain velocity, there's a different consideration that relates to the theory of relativity, that E^2 equals the rest energy squared plus the momentum, well the energy related to the momentum squared.

So p is the momentum, so it's mass times velocity, c is the speed of light, so this is momentum times speed of light, which is a similar type of quantity as mc^2 squared. For a photon, they have no mass, but this term still holds, so we can equate that to what we had already, E equals hf to get, well, pc equals hf , in other words the momentum of a photon is hf over c , so the energy of a photon scaled by the speed of light, and this is then the same as h over λ , because f is c over λ . So this is the momentum of a photon, and that momentum is a quantity that has a magnitude and a direction, and this holds in the wave propagation direction, so if light is traveling in that direction, then the momentum is also in that direction.

It is weird to wrap your head around this, something that does not have mass, still has momentum, but momentum can also be derived from a quantity that's energy. So if you scale the energy by the speed of light, you also get a quantity that is momentum, and that holds for photons, for things that have no mass.

So now get back to this one, so this was done in 1922 by Arthur and then it can interact with the electrons of that target, and then you get scattered photons at an angle ϕ with a different wavelength.

So to look at this from a physics point of view, we have an electron at rest that is hit by a photon with a certain momentum, then that a photon flies off at an angle ϕ with a different momentum, and then the electron goes from rest to velocity, and then it starts moving, also gets a momentum.

There's no time, so you only have conservation of momentum, so there is no force acting while this is happening, just momentum is conserved, so there's no impulse basically. Yes, they are, but if they move at a very low speed, much lower than the speed of light, the energy of an electron can be described just by this.

This is a really large quantity, m times c^2 . Only if it starts moving with a high velocity, this term starts to become relevant with respect to this one. So mainly, for all practical purposes, we have electrons at rest, unless you work with really, really high speed, and particle accelerators, they have to take this into account, not in the common world.

So the electron then recoils due to the impact, and this is what I said already, scattered photon flies off at an angle, and it has less energy and momentum, so it has a lower wavelength. And there is a way to do this, and that's derived from the book, I have a derivation here, but let me see if at the end of the lecture time permits, and I can show that as well. There's a nice derivation of this formula in the book, page 1291. What it comes down to is you do conservation of energy for this situation, and conservation of momentum.

So the momentum before, p , has to equal the momentum after, and since momentum is a vector quantity, you have a vector equation that you have to satisfy. All of these have to work out with their directions, and with their magnitudes. And this will lead to the formula by Compton, $\lambda' - \lambda$, so the wavelength of the scattered photon minus the wavelength of the original photon, so this is a positive value, this one is lower, and that one relates to Planck's constant, and the speed of light, and the mass, and then the angle at which the photon flies off.

So again here we need a particle description to describe this effect, a wave model cannot do this. So these are three arguments already why we need a quantized way of looking at light, so we need photons to describe this.

I think it's time for a break now, let's start again at half past, and then we'll go more in depth on how this works. So please take your seats, then we start again. Please take your seats, so let's see what you can make of this. One quiz question, and then we get to the really fun part.

Well, we don't have to be silent now, please discuss, see what you can make of this one.

I think the photoelectric effect has been covered in secondary school, but I'm not sure about Compton scattering and x-ray. So let's see what you can make of this.

So if you increase the number of electrons, what is going to happen?

Who thinks answer A? Answer B, quite a few, answer C, just one or two, answer D, answer E.

Answer B is indeed correct. By setting the voltage, you set the maximum amount of energy that can be converted into an x-ray, so you're setting the minimum wavelength.

If you increase the number of electrons you shoot at the anode, you just increase the intensity, not the frequency.

So there is a spectrum of frequencies you get out, but there is a certain minimum frequency, minimum wavelength that you can achieve. So let's get to the really fun part, as I said.

Quantum mechanics is really something that holds at the atomic level, so at really, really small scale.

There's Schrodinger's cat, of course, that tries to couple the really small scale to a microscopic example, but I'll get back to that in a second. This is really what I'm going to tell you now, it's really about what happens at a very small scale.

I'm not able to explain why this happens, I'm just going to tell you how it happens, and that's, well, that's the status quo in quantum physics. We know what happens, we don't know why it happens, we just can observe that it happens like that.

So I'll describe four experiments, and this is based on Feynman's lectures, which is a really nice read if you're interested. There are open source available on the internet, and these pictures are from that actually.

Suppose you have an experiment with a crappy gun, and you shoot that gun at a wall or metallic plate that has two slits in it, and what happens is that these bullets can ricochet off these gaps that you have in the wall, and then they can hit a backstop, or say a wooden wall in which they get stuck, and then you can count the bullets. I think we can agree that the bullet is something that is quantized, right? You can't have half a bullet, it's something that is just a unit of, well, propagation.

The thing that is shot through here. What happens if you do this is that if a bullet goes through hole one, it can ricochet off, or it can go directly through.

What happens afterwards is that if you would count the bullets and where they end up on the wall, you could develop a probability of finding the bullets somewhere like this. So it's most likely if it goes straight through to find it here, but if

it ricochets off the gap, you can, well, there will be spread on where you find the bullet, right? The same happens for slit two. You get P_2 as a probability density where you can find that bullet, and if you then combine the effects, so if the gun is really crappy, bullets can go through either of these holes, one bullet at a time per hole, and then the total probability would be the sum of these two. So the total probability of finding a bullet anywhere stuck in this wooden backstop would be this, the sum of the two individual probabilities. Yeah, go ahead.

Yeah, but these are, is a very sturdy metallic plate through which they are shot so they can really ricochet off the edge of the hole so they can then change direction. They can't, well, they can get back at you even, but then you don't find them in the wooden wall. But it's a crappy gun and a very good slit so they can hit the slit and then change direction, at least for a certain range.

So I guess we can agree that bullets arrive in lumps. It's a quantized thing and there's no interference here at the end. There's just a superposition of this probability or that probability, giving the total probability of where to find the bullet. Okay, example two.

If you have waves, so if you have, this is a top-down view of a wave pond and then there's a wave source that's an object bobbing up and down, generating circular waves, and if these waves go through slits, well, you get an interference pattern, just as we had with light. I'll get back to light later. But if you think of water waves, then each of these will cause a new source of wavelets and then you get an interference pattern on the back wall. And this can be an absorber, a beach of sand on which you detect where, what the wave height is at a certain moment.

The intensity, as we know for waves, relates to the magnitudes of the amplitude squared. So if the wave has a certain height and the power the wave carries relates to the height squared.

And if you have two sources, we can create an interference pattern. So there's local maxima, local minima, that you can detect on the absorber. So there's a local maximum here, there's probably a first minimum here, first minimum here, and then you get local maximum, minima, depending on where you are on the absorber.

So the intensity for each of these wave sources has a similar shape as what we had for the bullets, like the probability density for the bullets.

So these are similar types of, well, probabilities. But these intensities relate to the magnitude squared, the amplitude squared.

To create this interference pattern, you sum the amplitudes first, because that's how the waves interact, and then you look at the total power that is then left, and then you have the interference pattern. So it's not the sum of the two intensities, it's the sum of the amplitudes, which can be constructed for destructive interference, and then the power is the square of the sum of these amplitudes. That's how you create this interference pattern. This is also something we've seen before, right?

So waves are not quantized, they do not arrive in lumps, this amplitude of the wave can take any value, positive negative, bullets can't take negative values, right? The amplitude can take a negative value, and that's why the power relates to the amplitude squared. So this can range from zero to a certain maximum height, but as a continuous variable.

Okay, question? No, this is one by one. So after a certain amount of time, you've gathered a lot of bullets in the backstop here, and then if you look at where there are most bullets, if you just close one hole, you would find, well, probability p_1 ,

where there are most bullets stuck in the back wall.

How would the bullets interfere with another bullet? Okay, they could hit, yes. Yeah, so that's a difference between a lumped approach, bullets come in lumps, whereas waves are continuously, they propagate, and they have a wavefront, as we've defined, that arrives here, and then this creates two new wavefronts. So that's already a difference between the two types of interpretation. That's why one can give interference. So bullets are really fired one by one, whereas this is one wave source generating ultimately two new sources of waves, and then you can have interference.

I'll get back to photons and electrons in a second. But we can establish that the waves are not lumped, and they can show interference, right? Now the weird stuff happens.

If you have electrons, and you would fire them one by one, you can think of these as bullets. So I'm defining these to be fired one at a time, again, at a wall that has two slits, and then this detector could be, well, an electron detector, just a second, that gives a click when you hear an electron, right?

An electron detector plus an amplifier, you click, click, click when an electron ends up at the backstop. Question? I'm getting to that.

I'm getting to that.

This is one of the key concepts in quantum mechanics. This is a probability density function, and this is a probability.

So the density function now, let's consider the story that, okay, if you fire an electron through these slits, you can again get, well, a probability of finding an electron at a certain point on the back wall. If you would shoot them one at a time through a slit, you would find this pattern. If you would shoot them through, if you would close this slit and find them through that one, you would find this pattern.

The weird thing now for electrons is that the total pattern, if you would continue this for a while and then have some way of counting how much electrons have arrived where on the backstop, you would find an interference pattern.

Electrons are really discrete things here, quantized. They arrive in, well, it's a unit of, well, the smallest unit of charge, so they are lumped, but still they show interference pattern.

So next to being lumped, so next to thinking of them as particles, they also show behavior that is alike a wave, because they can show interference patterns.

So the probability of finding them is depicted here, and this relates to a probability density function, ϕ_1 , ϕ_2 , to the amplitude squared. But if they show interference, then we have to sum them in this way. So that's similar as what we did for the waves.

You can think of the probability density function as the predicted amplitude at a point in space, and the power or the intensity is then the square of that amplitude. And similar to here, the probability relates to the square of the intensity, of the probability density that you find an electron at some point on the back wall.

What you would hear in your detector is, if you would move that around and leave it there for some while, you would hear still the same click, so the same strength

every time, but you would just hear more clicks if you are at a region where there's a higher probability of finding electrons. And locally here, at this point, so here, there would be just a few clicks per every time instant, whereas at this point, just much more clicks per second.

So that's the rate at which you find the electrons. So the weird thing about quantum mechanics about, and I'll get to the weird thing about quantum mechanics, the weird thing about electrons is that they are lumped, but still show interference. That's something that really cannot connect.

But this is how it is in quantum mechanics. Well, let's try to be clever.

Let's use an electron gun and let's put a light source here, just after the first wall, and this light source will show when an electron comes past it. You will, if you're an observer here, you're watching this, you will see a brief flicker of light if an electron has gone this path. So you'll see a brief flicker of light here. If it has taken this path, it will show a brief flicker of light there. So you kind of see through which slit the electron has gone. If you set up your experiment like that, you will not find an interference pattern.

You will just find the sum of the two probabilities in the way that we had with the bullet experiment. And that's really weird. What has happened here is that your light source has interacted. So this is photons interacting with electrons. You can only see an electron if a photon has reflected off the electron.

So the light source has interacted with the electron going through this slit or that slit. And due to that interaction, it has changed the behavior of the electron. And once the behavior is changed, it does no longer show interference. So if you observe where they have come from, there is no interference anymore.

And that's the really weird part about quantum mechanics.

If you have them without observing what happens, you can have interference. And if you observe them or if you would close one of the slits, you would just get, if you would close slit two, you would just get P1 prime or vice versa. Close slit one, you would get P2 prime as the intensity of where you would find electrons. It would happen with light as well.

We're now talking electrons. This happens with light with photons as well. I'll come back to that in a later slide.

So this happens with electrons, which are the smallest particles we can look at. This also will happen to photons. And there's many interesting videos on that. I'll link a few in the slides as well.

So if you observe electrons, they show no interference. So depending on how you measure them, you're influencing them.

And also not nice from an experimental physics point of view. So you want to do accurate measurements, but you know that you're influencing the thing you're measuring by measuring it. So that's not nice.

So to try to consolidate what we've seen so far, these are the three principles of quantum whether you want it or not.

There is something called a probability amplitude, which relates to where we would find an electron if we would find that two slits in the example I showed. The probability is the square of this probability amplitude. Think of the probability amplitude as the wave amplitude, which can be positive or negative. And the

probability itself then relates to the intensity of the electrons ending up at a point in space. That's the square of the probability amplitude.

So this is that relation that we need to understand the concept.

If an event can occur in several alternative ways, so think of the electrons being able to go to slit one or slit two, then the resulting probability amplitude is the sum of the two individual probability amplitudes.

So think of the wave example again, the wave height caused by source one plus the wave height caused by source two, they sum. And if you then look at the intensity, you then square the resulting amplitude, which is then the probability.

And in that case, you can have interference because you sum before you square. And so that happens for the waves, but also happens for the electrons if there's two slits and you don't observe.

If there is an intermediate step in the experiment that helps determine which of the options is taken, so if there's multiple alternative ways, if you design your waves as taken place, then you don't have interference anymore. Then the resulting probability is just the sum of the individual probabilities, which relates to ϕ_1^2 plus ϕ_2^2 . So no longer the sum first and then squared, sorry, but then the individual probability amplitudes squared and then summed.

And this, well, the difference that you have, I'll get back to your question in a second, is if you think of the top products, ϕ_1^2 plus ϕ_2^2 gives ϕ_1^2 plus ϕ_2^2 plus the additional term that relates to the interference. And this is exactly the term that is missing. Well, that's the difference between these two considerations, between this description and this description.

There's this cross term which relates to the interference of the two terms, of the two sources, that is missing in this description and present in this description. What was your question?

Ah, so I'll state that this is complex now. I'll show you why it is complex on the lecture on Monday, because these are solutions to Schrodinger's equation, which is a complex valued wave equation. So solutions have to be complex. And that's why you had calculus, right?

I'm sorry.

This is unfortunately how quantum physics works. You need complex variables to describe this.

And it will follow from a solution to Schrodinger's equation.

This is the third Let me emphasize that better. So this would be a number three, just this part.

And this total thing would be number two. So the difference between the two is then this cross term, which relates to the interference.

So that's kind of a mathematical way of showing that this is different from this.

And exactly this cross term is what causes the interference. And if you are thinking complex variables, then this gets more complex by definition. So this is then the interaction between the two sources. But this term, this is just to show that there's a difference between summing first and then squaring and squaring the

sums. Your question.

No, it's a probability amplitude. Yeah, you can think of it like that. I'm not that into statistics to say if that is the variance or not. It might be.

I don't have a complete answer to that. Okay.

20 minutes. Good. So I gave an example for electrons.

This also holds if you would do photons one at a time. I would need a very accurate and very low power source to do this to shoot emit photons one at a time.

We know that with two slits, you get a two slit interference pattern. That's what we've seen in module six. If you would do this one photon at a time and you would have a photo resist material that would track where the photon has landed over time, you would build an interference pattern even if you do them one at a time. So then you're not talking about one photon interacting with one photon through slit one interacting with a photon through slit two. Even if you do them one at a time, you would still get this pattern. And that's the weird thing here. This is not, it's not possible to explain this. It's just something you have to assume that holds. It works like that. We're not sure why.

There have been many theories about hidden variables describing this.

So there's an underlying hidden mechanism that describes how these photons react. So that if it goes through slit one, it will do this. If it goes through slit two, it will do that. But these are not true. These have been all been proven false. Well, I'm the one I'm teaching.

There is many interpretations on how this works.

I have learned to accept that it works like this. And I'm trying to teach you the same concept without knowing exactly why it works like this. It is just like that. I'm sorry.

So sometimes you have to take things for granted.

This is one of them. So there was a friend guy, de Broglie, who predicted, well, hypothesized that electromagnetic waves can behave as particles. And he also, well, he hypothesized actually the opposite. Can particles behave as waves? So is duality. If a wave can behave like a particle, can it be done the other way around? So and he proposed towards the end of the development of quantum mechanics that the motion of a particle is governed by an associated matter wave that has a certain frequency and wavelength that relates to the momentum.

So what he proposed was the following. So the formula we've seen before, P equals $\hbar$ over λ .

That's what he hypothesized. So the momentum of a particle, whatever particle, relates to Planck's constant and the wavelength. So even for microscopic things, this holds. But we are, since Planck's constant is really, really small, we end up with weird numbers that are not interpretable. This can also be written as $\hbar$ times K , where K is the wave number. K was 2π over λ , over λ .

So if we would put that in, $\hbar$ Let me see. I'm doing this correctly. So I'm 2π over λ .

I can write this as K times a scaled Planck's constant $\hbar$, which is $\hbar$ over 2π . So this is just a matter of terminology. Sometimes in books you find $\hbar$, sometimes

you find $\hbar$. The difference is the scaling factor of 2π .

This is how it's used. And in Schrodinger's equations, a lot of π 's will appear because of this, basically. So De Broglie hypothesized that every particle that has a certain energy can be associated with a wave that has a certain wavelength. And to show how this works, there's proof of that, actually.

So these are two images, two half-images, basically, of the top image showing x-rays of a wavelength of 71 picometers passing through aluminum foil.

These give a diffraction pattern. So a circular diffraction pattern which has one center-bright circle and then rings going further out. And if you shoot electrons at aluminum, so we're talking about electromagnetic waves, light, and electrons, you would get a similar diffraction pattern. So this confirms that the wave behavior is correct.

So these are different scales, but they still give a similar type of diffraction pattern.

So the tricky thing about quantum mechanics is that instead of having a deterministic description, we now have to use probability amplitudes. So we don't have complete predictability anymore.

Einstein really did not like this. He said, God does not play dice. So until his death, he could not live with that idea, but still he helped formulate that idea in the end. That's a start, actually. What it comes down to is that we have probabilities, we have particles and waves that can be described by the same formalism, but never at the same time.

When you want to explain something, you either choose the particle interpretation or the wave interpretation.

You need them both, but never at the same time. And that's called the complementarity principle, founded by Niels Bohr, also one of the fathers of quantum physics.

But you need them both, just never at the same time.

And there's a very, very nice video of Veritasium.

You probably know that channel as well.

Check this out. It's too long to show, but I've linked it.

I want to show a part of it. Check this out. I'm using this speaker to vibrate a Petri dish containing silicon oil. Now, if I take this, the droplet is actually bouncing and it will keep bouncing for a very long time. Now, the reason for this is there's a little layer of air between the droplet and the surface. And the droplet is bouncing so rapidly that that layer never shrinks to about 100 nanometers, which is what it would take for the droplet to recombine with the oil. Now, every time the droplet lands on the surface, it creates a wave. But this is a special type of wave driven by the vibration of the oil bath. It is a standing wave, meaning that it's not traveling out, it's just oscillating up and down. So the droplet makes the wave and then it interacts with that wave on its next bounce. If the drop lands on one side of the wave, it is pushed forwards. And as long as the bounce of the droplet remains synchronized with the wave, it'll keep landing on the front side of the wave and getting pushed farther forwards. Droplets like these are known as rockers. The bouncing oil drops have been known about since the 1970s, but only recently has it been discovered that you can use these little droplets to

replicate many of the strange phenomena of quantum mechanics. Now, obviously, this is not a quantum system. The droplets are about a millimeter in diameter. But you can think of the droplets like quantum particles, say electrons.

One experiment that captures the key features of quantum mechanics is the double slit experiment. If you send a beam of electrons at two narrow slits, the electrons, rather than behaving like particles and ending up in two clumps behind the slits, they produce an interference pattern, even when you send each electron through one at a time. With walking droplets, the pilot wave goes through both slits, interfering with itself, while the droplet only goes through one slit. The droplet doesn't move in a straight line, though. It's deflected by its interaction with the wave. The resulting distribution of where the droplets end up looks very similar to quantum double slit interference patterns, or take tunneling. In quantum mechanics, it's possible for a particle to get through a barrier that it wouldn't classically have enough energy to get over. This has been demonstrated with walkers by creating a shallow barrier under the surface of the oil. Usually, the barrier reflects the pilot wave and its bouncing droplet, but in rare cases, the droplet does cross the boundary. And the probability of the droplet crossing the barrier decreases exponentially with the increasing width of the barrier, just as in quantum tunnel.

So this is, please watch the video if you're interested. This is really a, as he said, it's not a quantum system. It's too large for that. Quantum effects really happen at the atomic scale. But this is one way of trying to understand how a particle also relates to wave behavior and how a wave can influence the movement of that particle itself when it goes through the slits. Very nice, very nice video.

Another way of connecting microscopic things to microscopic things is Schrodinger's cat. That's how I remember the t-shirt.

If you think about a cat that is sealed in a room, the cat has food, so it can stay alive if it wants to. I'm not sure what cats want. And there is a radioactive element that may or may not decay. So depending on whether it's a radioactive element, there's a half time, so it may decay. And if it decays, then a Geiger counter measures that and it will shatter a flask of poison that may kill the cat. And as long as you don't open the box, you don't know what has happened. So you can think of the cat as being in a superposition of two states, being alive and being dead at the same time. And this relates to the probability of things that may or may not have happened. Only opening the box during a measurement can answer that question. Similarly, your midterm test results are still in this quantum superposition of you having passed the midterm and you having failed the midterm. We are currently working hard on the measurements, but we'll get back to that next week. I think in view of time, let me skip this.

We had a lot of questions and interaction already. There's still a few slides I want to cover. Other examples of daily life. I would be worried if I found this in my fridge. It's one way of protecting your objects. Schrodinger's leftovers, which can be good, bad, or both. Please try it in your student home.

There's also more philosophical takes on this. There's the many worlds interpretation that states that every time at which there is a constant superposition of states, then as soon as you make a measurement, you continue on one branch, but the other branch still exists. You just have not selected that branch. There is also some critique on this because this does not take into account the weighting between the different states that you have. It's just a branch which looks like a 50-50 chance, but this is one interpretation that is possible. This is also in popular science, of course. You know Rick and Morty probably. This was a very nice episode on all the different states that could happen one Morty time-traveled.

He got a lot of trouble there.

To conclude today's lecture, inherent to this is, well inherent to quantum mechanics, is that there is probability.

There's also uncertainty.

You can't do measurements with absolute certainty.

That relates to Heisenberg's uncertainty principle.

If you look at individual particles, so let's get back to the photons that go through.

This is the two-slit case. This does not match this picture actually, but in the two-slit case, you could have them end up anywhere on the screen. Only if you do sufficient, sufficiently many, you would find an interference pattern like this. So the behavior of the large ensemble is predictable, but the individual particles, you don't know where they are with absolute certainty.

For some combinations of measurements, there is a fundamental limit to the accuracy that you can achieve.

The most well-known example, that's what Heisenberg developed, although he wasn't sure at the time. Sorry for that.

He connected the position of the particle you would measure to the momentum of the particle that you would find, and that relates to where it would end up on the screen. So depending on where you would find, if you would do a really, really accurate measurement of the particle, then you would have a large uncertainty in the momentum.

And the reasoning behind this, well, this also happens for electrons. Let me get my train of thought. Yeah, this also happens for electrons that have wave-like behavior if you have many electrons, but the particle-like behavior becomes evident if you have only a few electrons. So this is a similar picture as for photons, but now with electrons shot one at a time through this double slit. So this is really the experiment that Feynman described as well.

After many electrons, you will get such an interference pattern, but we cannot predict for a single electron where it will end up. Just the ensemble, you can predict.

So we may know where they are, but not precisely where they are going or vice versa. And how Heisenberg derived this was kind of by a thought experiment, and you can relate that to classical optics. If you would have a gamma photon incident on an electron, there would be scattering, and then based on where you would have the electron, there would be a range in angles that you could achieve, and that relates to the resolution in classical optics. It's fine if you can follow the main line. What matters is the result on the next slide. The resolution you would have here relates to the angle you would find, and if you look at the momentum that a photon has, then that also has, well, this photon also has the sine of the angle in there. And if you then take the product of the two, you end up with something that is proportional to Planck's constant.

So the product of Δx times Δp has the sines cancel, the lambdas cancel, so this is proportional to Planck's constant. And this provides a fundamental lower bound to things we can measure. A sharp lower bound.

You cannot. Well, it was proportional to, in the previous slide, if you would do the math, you would find that you have $\hbar$ over 2 as being the fundamental limit that you have for Δx times Δp . So if you have a Δx versus a Δp plot, there is a certain lower bound below which you cannot go. You either measure the position accurately, and you don't know about the momentum, or vice versa, or you can choose a combination that, well, sacrifices both of these.

And the deeper cause behind this, and that also relates to, well, the experiment I described with the light source after the two slits, is your measurements influence the state of the particle, or the wave, or both. And the measurement results does depend on the order of measurements. So that's really a fundamental concept that is, well, inherent to quantum mechanics. So this is the probably most well-known one, position and momentum uncertainty. So we cannot simultaneously know the position and the momentum of a photon really well, or any, or an electron, for that matter. But there's also a principle for energy and time, the better you know the energy of a photon, the less certain you will know when to observe the photon. Or the better you know what energy level an atom is in, the less sure you can predict at what time this electron will go to a different energy state.

Because also there, there's a fundamental uncertainty due to Heisenberg.

It's time already, this. Two quiz questions, actually.

Yeah, I took more time than I expected for the Feynman thing, but which was nice, I hope. Let's do the quiz questions, and then we'll call it a day. So probably the uncertainty principle was covered in secondary school, right? And if you know the principle, this is kind of a no-brainer, hopefully. Who thinks answer A?

Answer B? Answer C? Answer D? Yeah, indeed.

Answer D is correct because it scales like inversely proportional. This one is a bit more tricky, and then we're done.

What can you tell about a photon that should land in the center of that diffraction pattern?

And this is a one-slit diffraction pattern.

Who thinks answer A? Answer B? Answer C? Answer D?

Answer E? Yeah, it's not either of the two. You don't know where it has gone through because that's the uncertainty you have inherently in quantum mechanics.

It's impossible to predict what happens to an individual particle. So further reading or further watching, rather, is another nice video here. And then I'll see you on Monday. Have a nice weekend already.